

Scenario 1: Student Performance Analyzer

Use multiple standard-library modules to build a small student performance analyzer.

Requirements

Given student marks:

```
students = {"Tarun": [82, 91, 78], "Karthik": [65, 72, 68], "Chadrika": [91, 95, 89], "Trisha": [76, 81, 79], "Atul": [88, 84, 92]}
```

The program should:

- Calculate each student's average.
- Find the highest average.
- Randomly select one student.
- Display the current date.
- Use Counter to classify grades.
- Use reduce() to calculate total marks.

Scenario 2: Automated Course Project Team Allocation

Imagine that your **MCA department** is organizing a semester-end **industry-oriented project**.

There are 12 students from different departments:

- CSE students
- Data Science students
- AI students
- Cyber Security students

The department wants to automatically create project teams instead of manually assigning students.

Real-world situation

The coordinator enters the required team size from the command line:

```
python allocation.py 3
```

Here, 3 means **each project team should contain 3 students**.

The program then performs the following operations:

1. **Group students by department**
 - CSE students are grouped together.
 - AI students are grouped together.
 - Data Science students are grouped together.
 - This can be handled using defaultdict.
2. **Generate possible teams**
 - The program generates different combinations of students.
 - For example, from 4 students:
 - (Anita, Rahul, Priya)
 - (Anita, Rahul, Kiran)
 - (Anita, Priya, Kiran)

- (Rahul, Priya, Kiran)
- `itertools.combinations()` is useful here.
- 3. **Randomly select a team**
 - Instead of always selecting the first combination, the system randomly chooses a team.
 - `random.sample()` can be used.
- 4. **Calculate the team's total marks**
 - Suppose the selected students have marks:
 - Anita → 85
 - Rahul → 78
 - Priya → 92
 - Total: $85 + 78 + 92 = 255$
 - `functools.reduce()` performs this calculation.
- 5. **Calculate average marks**
 - Using the total and number of students: $\text{Average} = 255 / 3 = 85$
 - `math` can be used for mathematical processing.
- 6. **Record the allocation date**
 - The system records when the team was created:
Allocation Date: 25-09-2026
 - `datetime` is used.
- 7. **Accept team size from the command line**
 - Instead of modifying the program every time, the coordinator enters: `python allocation.py 3`
 - `sys.argv` retrieves 3.

Example final output

===== PROJECT TEAM ALLOCATION =====

Team Size: 3

Allocation Date: 25-09-2026

Department: CSE

Selected Team:

1. Anita - 85

2. Rahul - 78

3. Priya - 92

Total Marks: 255

Average Marks: 85.00